9 Mr Tan is looking to purchase a new car to drive to and from work on weekdays and for leisure on weekends. He estimates the total distance travelled is about 1500 km per month. The average cost of petrol is \$2.20 per litre. To buy a new car, he must pay a down payment of 30% of the selling price of the car before he can take a car loan for the remaining amount from a bank. He has to pay back the loan by monthly instalment. A 5-year car loan simple interest rate offered by most banks is 2.28% per annum. Mr Tan shortlisted 3 cars with all the relevant cost as shown in the table.

	Car A	Car B	Car C
Selling Price	\$87 999	\$108 999	\$107 888
Fuel Consumption(km per litre)	17.2	14.9	17.8
Car Insurance per year	\$ 1 200	\$ 1500	\$ 1 500
Engine Capacity(in cubic cm)	1 598	1 499	1 197
Monthly Car Maintenance	\$200	\$200	\$200
Monthly Car Park charges	\$150	\$150	\$150

He also finds out that the annual road tax of the car is determined by the engine capacity of the car is as follows:

$$\text{Annual Road Tax} = [\$500 + 0.75(\text{Engine Capacity minus } 1000)] \times 0.782$$

(a) After reviewing the information, Mr Tan decides to buy Car A.

(i) calculate the down payment.

Answer \$.. [1]

(ii) calculate the total **monthly** expenditure including monthly instalment payment and all the other monthly costs.

Answer \$.. [7]

(b) An alternative to buying a car is to rent an electric car that is easily accessible from his house and workplace. Subscription per month is \$15 and it costs 33 cents per minute of use. Mr Tan estimates that the average daily travel time to and from work is about 1 hour and 20 minutes. On weekends, he needs to use the car for about 5 hours for leisure activities. Calculate his monthly expenditure to rent a car.

Answer \$…………………………………….. [2]

(c) Do you think Mr Tan should buy or rent a car? Justify your answer.

Answer………………………………………………………………….………………………

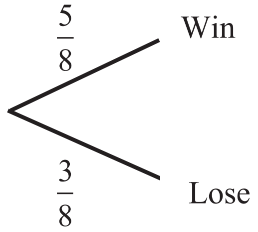
………..…………………………………………………………………….…………..[1]

10 **(a)** Team Alpha and Delta are competing in the badminton finals. Each game will only result in a win or a loss. The competition ends when a team wins 2 games out of 3.

The probability of Alpha team winning in a game is $\frac{5}{8}$.

(i) Draw a tree diagram to show all the possible outcomes for Alpha team.

Answer



[2]

(ii) Calculate the probability, expressing your answers in fraction, that Alpha team wins the competition.

Answer …………………………………….. [2]

10 **(b)** The frequency table shows the weight in kg of 80 persons who join an exercise club.

Weight	$30 < x \leq 40$	$40 < x \leq 50$	$50 < x \leq 60$	$60 < x \leq 70$	$70 < x \leq 80$
Frequency	8	17	34	18	3

(i) Find the mean weight.

Answer ………………………kg [1]

(ii) Find the standard deviation.

Answer ………………………..kg [1]